Prodigy Anesthesia Quick Review

Acidosis

What is the definition of metabolic acidosis?	Metabolic acidosis is caused by the presence of a metabolic acid, which lowers the pH to less than 7.35 *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 372-373.*
What is the definition of respiratory acidosis?	Respiratory acidosis occurs when alveolar ventilation decreases enough that the PaCO2 rises, leading to a pH less than 7.35 *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 372.*
What are the major cardiac effects of acidosis and at what pH does this occur?	Acidosis decreases myocardial contractility. The effects are usually not clinically evident until the pH is less than 7.2 when the ability of the heart to respond to catecholamines is decreased. It is more evident, however, in patients with impaired left ventricular function, those on beta-blockers, or patients under general anesthesia. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 370-371.*
A patient is suffering from an exacerbation of chronic ventilatory failure and exhibits severe acidosis. You are about to intubate the patient and place him on the ventilator. Should the acidosis be corrected urgently or slowly? Why?	Chronic respiratory acidosis should not be corrected too rapidly because the lungs are able to eliminate CO2 much more quickly than the kidneys can eliminate bicarbonate. If the CO2 is eliminated too quickly, the patient will suffer from a metabolic alkalosis that can lead to neuromuscular irritability and CNS excitation that can lead to seizures. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 372.*
What is the most likely cause of respiratory depression in the perioperative period?	Drug-induced depression of ventilation due to opioids or general anesthetics *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 372.*
What are the primary causes of respiratory acidosis?	The primary causes of respiratory acidosis can be divided into two categories: those due to alveolar hypoventilation and those due to increased CO2 production. Factors that impair CO2 elimination such as pulmonary diseases, neuromuscular disorders, chest wall abnormalities, and obtundation can potentially result in alveolar hypoventilation. Factors such as intense shivering, malignant hyperthermia, prolonged seizure activity, thyroid storm, and extensive burns result in an increase in CO2 production. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1149.*

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A patient with metabolic acidosis has a sodium of 148 mEq/L, a chloride of 103 mEq/L, and a bicarbonate level of 19 mEq/L. Is this a normal anion gap acidosis or a high anion gap acidosis?	The formula for the anion gap is: [Na+] - ([Cl-] + [HCO3-]). In this instance, the anion gap is 26 mEq/L. The normal anion gap is 7 to 12 mEq/L, so this is a high anion gap acidosis. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1150-1152.*
What are the two major classifications of metabolic acidosis?	Normal anion gap acidosis and high anion gap acidosis *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 372-373.*
What systemic abnormalities may occur with acidosis?	Severe acidosis is associated with decreased cardiac contractility, decreased responsiveness to catecholamines, sensitization to re-entrant tachydysrhythmias, lowered threshold for ventricular fibrillation, hyperkalemia, hyperventilation, insulin resistance, and inhibition of anaerobic glycolysis. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 372-373.*
How does acidosis affect the oxyhemoglobin dissociation curve?	The oxyhemoglobin dissociation curve shifts to the right in acidosis to allow for a greater unloading of oxygen from the hemoglobin in the tissue bed. Other factors that shift the oxyhemoglobin dissociation curve to the right include: increased 2,3 DPG, fever, elevated CO2, and low carbon monoxide levels. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1149.*
How does high anion gap acidosis occur?	A high anion gap acidosis occurs when a fixed acid is introduced into the extracellular space. As the acid dissociates, the hydrogen ion bonds with bicarbonate to form carbonic acid. It is the drop in bicarbonate that produces the anion gap. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 372-373.*
What are the most common causes of a normal anion gap acidosis?	Conditions that result in an increased production of nonvolatile acids such as renal failure, ketoacidosis, nonketotic hyperosmolar coma, and rhabdomyolysis or the ingestion of toxins such as salicylates, methanol, paraldehyde, and ethylene glycol can result in an acidosis with an increased anion gap. Acidosis with a normal anion gap can be produced by increased gastrointestinal losses of HCO3- from diarrhea, fistulas, or ingestion of CaCl2 or MgCl2. Other factors that can produce a normal anion gap acidosis include hypoaldosteronism, renal tubular acidosis, carbonic anhydrase inhibitors or the administration of large volumes of bicarbonate-free intravenous fluids. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1151.*

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What is another name for normal anion gap acidosis?	Hyperchloremic metabolic acidosis *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 372.*
What are the characteristics of normal anion gap acidosis?	Normal anion gap acidosis is due to an increase in chloride concentration. This usually occurs in instances where bicarbonate is lost, so the kidneys retain chloride ions to maintain electrical neutrality. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 372-373.*

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